

5. Categorical Cross Entropy [used in Softmax Regression]

→ [Multi-class] {classification}

1 point →
$$L = - \sum_{j=1}^K y_j \log(\hat{y}_j)$$

(K = no. of classes)

$$L = -y_1 \log(\hat{y}_1) - y_2 \log(\hat{y}_2) - y_3 \log(\hat{y}_3)$$

for yes.

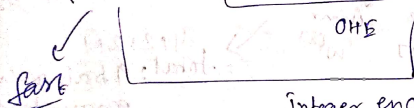
$$f(z) = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

Activation
Softmax

for no.

$$\frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}} \quad \text{for Maybe} \rightarrow \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

Sparse categorical cross entropy



Integer encode.

[0.1, 0.4, 0.5] → 1.

Suppose $(y_1, y_2, y_3 = 1)$

① → (7, 70) → $-1 \times \log(0.1)$

[0.1, 0.4, 0.6] → 2 → $L \rightarrow (8, 80)$

$-1 \times \log(0.4)$

[0.1, 0.4, 0.6] → 3 → $L \rightarrow (6, 60)$

$-1 \times \log(0.6)$

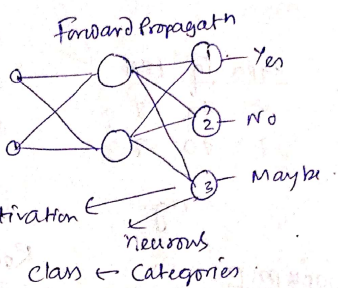
$$L = - \sum_{j=1}^K y_j \log(\hat{y}_j) \rightarrow C = - \frac{1}{n} \sum_{j=1}^n \sum_{i=1}^K y_{ij} \log(\hat{y}_{ij})$$

Summary.

Key → MSE
→ outliers → MAE
→ huber loss

Classification → binary → Binary class classification (BCE)
→ Multi → few cat → Categorical Cross Entropy (CCE)
→ many " → (SCE) → sparse

cgpa	iq	placed	Yes	No	Maybe
8	80	Yes	1	0	0
6	60	No	0	1	0
7	70	Maybe	0	0	1



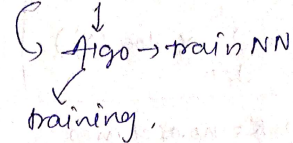
cgpa	iq	placed	Placed
7	70	Yes	1
8	80	No	2
6	60	Maybe	3

Backpropagation

Prerequisites

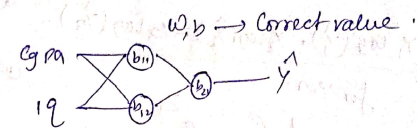
→ Gradient descent
→ forward propagation

Backprop



cgpa	iq	LPA
8	80	8
7	70	7

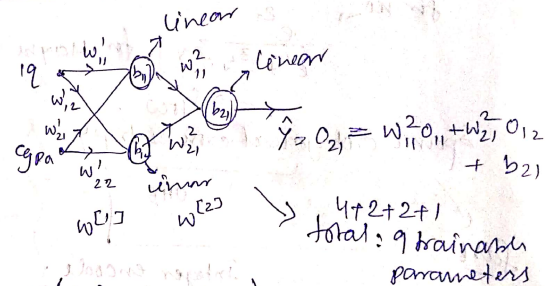
reg.



backprop.

iq	cgpa	LPA
80	8	3
60	9	5
70	5	8
120	7	11

Regression



Steps: 0) init w, b (let w=1, b=0)

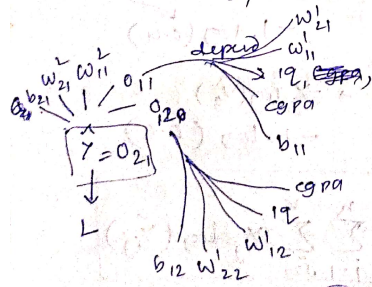
1) you select a pt. (row)

2) Predict (LPA) → forward propagation [Dot product]

3) choose a loss fn. → MSE

4) w, b - update

→ Gradient descent



$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w_{old}}$$

$$b_{new} = b_{old} - \eta \frac{\partial L}{\partial b_{old}}$$

$$w_{11}^2_{new} = w_{11}^2_{old} - \eta \left(\frac{\partial L}{\partial w_{11}^2_{old}} \right)$$

derivative of loss.

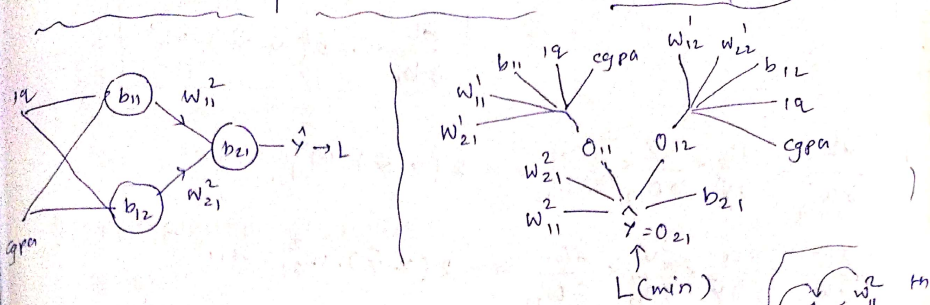
$$w_{21}^2_{new} = w_{21}^2_{old} - \eta \frac{\partial L}{\partial w_{21}^2_{old}}$$

→ 9 trainable parameters

$$b_{21}_{new} = b_{21}_{old} - \eta \frac{\partial L}{\partial b_{21}_{old}}$$

→ 9 derivatives ✓

$$\left[\frac{\partial L}{\partial w_{11}^2}, \frac{\partial L}{\partial w_{21}^2}, \frac{\partial L}{\partial b_{21}} \right] \quad \left[\frac{\partial L}{\partial w_{11}^1}, \frac{\partial L}{\partial w_{21}^1}, \frac{\partial L}{\partial b_{11}} \right] \quad \left[\frac{\partial L}{\partial w_{12}^1}, \frac{\partial L}{\partial w_{22}^1}, \frac{\partial L}{\partial b_{12}} \right]$$



You cannot find $\frac{\partial L}{\partial w_{11}^2}$ directly. !!

You have to find out $\rightarrow \frac{\partial L}{\partial w_{11}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_{11}^2}$ (indirect change)

$$\frac{\partial L}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}} (\hat{y} - y)^2 = -2(\hat{y} - y)$$

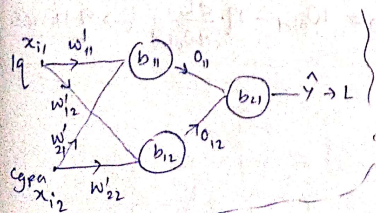
$$\frac{\partial \hat{y}}{\partial w_{11}^2} = \frac{\partial}{\partial w_{11}^2} [0_{11} w_{11}^2 + 0_{12} w_{21}^2 + b_{21}] = 0_{11}$$

$$\therefore \frac{\partial L}{\partial w_{11}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_{11}^2} = -2(\hat{y} - y) \cdot 0_{11} \quad \left[\frac{\partial L}{\partial w_{11}^2} = -2(\hat{y} - y) \cdot 0_{11} \right]$$

$$\frac{\partial L}{\partial w_{21}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_{21}^2} = -2(\hat{y} - y) \cdot 0_{12} \quad \left[\frac{\partial L}{\partial w_{21}^2} = -2(\hat{y} - y) \cdot 0_{12} \right]$$

$$\frac{\partial L}{\partial b_{21}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial b_{21}} = -2(\hat{y} - y) \cdot 1 = -2(\hat{y} - y) \quad \left[\frac{\partial L}{\partial b_{21}} = -2(\hat{y} - y) \right]$$

$$\frac{\partial L}{\partial w_{11}^1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial 0_{11}} \times \frac{\partial 0_{11}}{\partial w_{11}^1} = -2(\hat{y} - y) \cdot w_{11}^2 \cdot x_{11} \quad \left\{ \begin{array}{l} 0_{11} = x_{11} w_{11}^1 + x_{12} w_{21}^1 + b_{11} \\ 0_{12} = x_{11} w_{12}^1 + x_{12} w_{22}^1 + b_{12} \end{array} \right.$$



$$\frac{\partial L}{\partial w_{21}^1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial 0_{12}} \times \frac{\partial 0_{12}}{\partial w_{21}^1} = -2(\hat{y} - y) \cdot w_{21}^2 \cdot x_{12}$$

$$\frac{\partial L}{\partial w_{11}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial 0_{11}} \times \frac{\partial 0_{11}}{\partial w_{11}^2} = -2(\hat{y} - y) \cdot w_{11}^2 \cdot 1$$

$$\left[\frac{\partial L}{\partial w_{11}^1} = -2(\hat{y} - y) \cdot w_{11}^2 \cdot x_{11} \right] \quad \left[\frac{\partial L}{\partial w_{21}^1} = -2(\hat{y} - y) \cdot w_{21}^2 \cdot x_{12} \right]$$

$$\left[\frac{\partial L}{\partial b_{11}} = -2(\hat{y} - y) \cdot w_{11}^2 \right]$$

$$\frac{\partial L}{\partial w_{12}^1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial 0_{12}} \times \frac{\partial 0_{12}}{\partial w_{12}^1} = -2(\hat{y} - y) \cdot w_{21}^2 \cdot x_{12}$$

$$\frac{\partial L}{\partial w_{22}^1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial 0_{12}} \times \frac{\partial 0_{12}}{\partial w_{22}^1} = -2(\hat{y} - y) \cdot w_{21}^2 \cdot x_{12}$$

$$\frac{\partial L}{\partial b_{12}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial 0_{12}} \times \frac{\partial 0_{12}}{\partial b_{12}} = -2(\hat{y} - y) \cdot w_{21}^2$$

$$\left[\frac{\partial L}{\partial w_{12}^1} = -2(\hat{y} - y) \cdot w_{21}^2 \cdot x_{12} \right] \quad \left[\frac{\partial L}{\partial w_{22}^1} = -2(\hat{y} - y) \cdot w_{21}^2 \cdot x_{12} \right]$$

$$\left[\frac{\partial L}{\partial b_{12}} = -2(\hat{y} - y) \cdot w_{21}^2 \right]$$

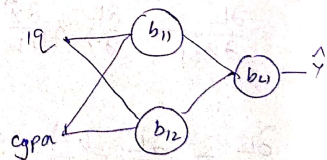
Step 1: Weigh, bias $\rightarrow$ random $\rightarrow$ random
for i in range(4):
1) a: $\pm$ student $\rightarrow$ forward propagation

predict (LPA)

2) b: loss calculate (MSE)

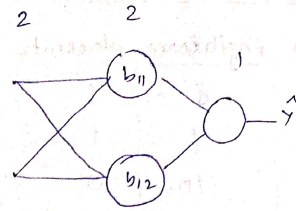
3) c: Adjust all weights and biases (new w, b)

$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w_{\text{old}}} \rightarrow \text{loop 9-times}$$



Regression Problem

gpa	profile score	placement
8	8	4
7	9	5
6	10	6
5	12	7



Back propagation Algorithm

epochs = 5

for i in range(epochs):

for j in range(X.shape[0]):

- select 1 row (random)
- predict (using forward propagation)
- calculate loss (using loss fn → mse)
- update weights & bias using GD

$$w_n = w_0 - \eta \frac{\partial L}{\partial w_0}$$

→ calculate avg loss for the epoch

$$\begin{aligned} \frac{\partial L}{\partial w_{11}^2} &= -2(y - \hat{y}) \cdot 0_{11} & \frac{\partial L}{\partial w_{11}^1} &= -2(y - \hat{y}) w_{11}^2 x_{i1} & \frac{\partial L}{\partial w_{12}^1} &= -2(y - \hat{y}) w_{12}^2 x_{i1} \\ \frac{\partial L}{\partial w_{21}^2} &= -2(y - \hat{y}) \cdot 0_{12} & \frac{\partial L}{\partial b_{11}} &= -2(y - \hat{y}) w_{11}^2 & \frac{\partial L}{\partial w_{22}^1} &= -2(y - \hat{y}) w_{22}^2 x_{i2} \\ \frac{\partial L}{\partial b_{21}} &= -2(y - \hat{y}) & \frac{\partial L}{\partial w_{21}^1} &= -2(y - \hat{y}) w_{21}^2 x_{i2} & \frac{\partial L}{\partial b_{12}} &= -2(y - \hat{y}) w_{21}^2 \end{aligned}$$

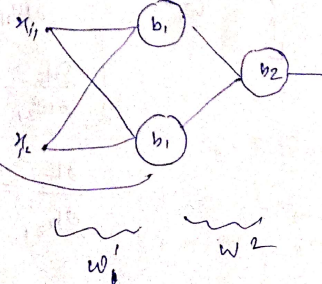
Regression Problem

$$b_1 = [-, -]$$

$$b_2 = [-]$$

$$w^1 = \begin{bmatrix} - & - \\ - & - \end{bmatrix}$$

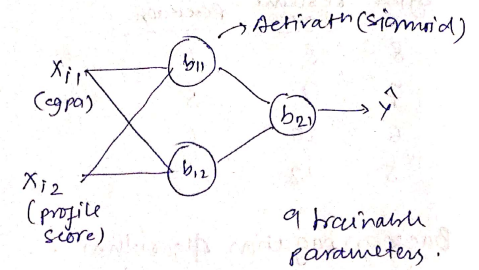
$$w^2 = \begin{bmatrix} - \\ - \end{bmatrix}$$



Classification Example

21/07/25

gpa	profile score	placement
8	8	1
7	9	1
6	10	0
5	5	0



Back propagation algorithm

epochs = 5

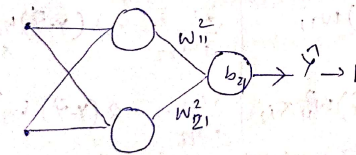
for i in range(epochs):

for j in range(X.shape[0]):

- select 1 row (random)
- predict (using forward prop)
- calculate loss (using loss function → bce)
- update w, b using GD

$$w_n = w_0 - \eta \cdot \frac{\partial L}{\partial w_0} \rightarrow g_{w,b}$$

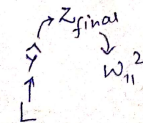
→ Calculate avg loss for the epoch



$$\hat{y} = \sigma(z)$$

$$L = -y \log(\hat{y}) - (1-y) \log(1-\hat{y})$$

$$z_{final} = w_{11}^2 \cdot 0_{11} + w_{21}^2 \cdot 0_{12} + b_{21}$$



$$\frac{\partial L}{\partial w_{11}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial w_{11}^2}$$

$$\frac{\partial L}{\partial w_{21}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial w_{21}^2}$$

$$\frac{\partial L}{\partial b_{21}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial b_{21}}$$

$$\frac{\partial L}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}} [-y \log(\hat{y}) - (1-y) \log(1-\hat{y})]$$

$$= -\frac{y}{\hat{y}} + \frac{(1-y)}{(1-\hat{y})} (-1) = \frac{-y(1-\hat{y}) + \hat{y}(1-y)}{\hat{y}(1-\hat{y})}$$

$$= \frac{-y + y\hat{y} + \hat{y} - y\hat{y}}{\hat{y}(1-\hat{y})}$$

$$\hat{y} = \sigma(z)$$

$$= \frac{(\hat{y}-\hat{y})}{\hat{y}(1-\hat{y})}$$

$$\frac{\partial \hat{y}}{\partial z} = \frac{\partial}{\partial z} \sigma(z) = \sigma(z) [1 - \sigma(z)]$$

$$= \hat{y}(1-\hat{y})$$

$$\frac{\partial L}{\partial \hat{y}} = -\frac{(y-\hat{y})}{\hat{y}(1-\hat{y})} \cdot \frac{\partial \hat{y}}{\partial z} = -\frac{(y-\hat{y})}{\hat{y}(1-\hat{y})} \cdot \hat{y}(1-\hat{y})$$

/ multiply

$$\left\{ \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} = -(y-\hat{y}) \right\}$$

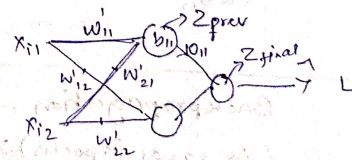
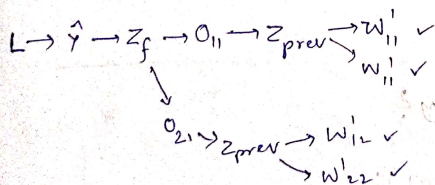
$$\frac{\partial z}{\partial w_{11}^2} = o_{11}$$

$$\therefore \frac{\partial L}{\partial w_{11}^2} = -(y-\hat{y}) \cdot o_{11}$$

$$\frac{\partial L}{\partial w_{21}^2} = -(y-\hat{y}) \cdot o_{12}$$

$$\left\{ z = w_{11}^2 o_{11} + w_{21}^2 o_{12} + b_{z1} \right\}$$

$$\frac{\partial L}{\partial b_{z1}} = -(y-\hat{y})$$



$$\frac{\partial L}{\partial w_{11}^1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_f} \times \frac{\partial z_f}{\partial o_{11}} \times \frac{\partial o_{11}}{\partial z_{prev}} \times \frac{\partial z_{prev}}{\partial w_{11}^1}$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$-(y-\hat{y}) \quad w_{11}^2 \quad o_{11}(1-o_{11}) \quad x_{i1}$$

$$\left\{ \begin{aligned} o_{11} &= \sigma(z_{prev}) \\ \frac{\partial o_{11}}{\partial z_{prev}} &= \sigma'(z_{prev}) \\ &= o_{11}(1-o_{11}) \end{aligned} \right.$$

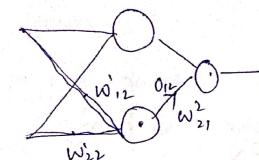
$$z_{prev} = x_{i1} w_{11}^1 + x_{i2} w_{21}^1 + b_{11}$$

$$\frac{\partial z_{prev}}{\partial w_{11}^1} = x_{i1}$$

$$\frac{\partial L}{\partial w_{11}^1} = -(y-\hat{y}) \cdot w_{11}^2 \cdot o_{11} \cdot (1-o_{11}) \cdot x_{i1}$$

$$\frac{\partial L}{\partial w_{21}^1} = -(y-\hat{y}) \cdot w_{11}^2 \cdot o_{11} \cdot (1-o_{11}) \cdot x_{i2}$$

$$\frac{\partial L}{\partial b_{11}} = -(y-\hat{y}) \cdot w_{11}^2 \cdot o_{11} \cdot (1-o_{11})$$



$$\frac{\partial L}{\partial w_{12}^1}, \frac{\partial L}{\partial w_{22}^1}, \frac{\partial L}{\partial b_{12}}$$

$$L \rightarrow \hat{y} \rightarrow z_f \rightarrow o_{12}$$

$$w_{12}^1 \leftarrow z_p$$

$$\frac{\partial L}{\partial w_{12}^1} = \left\{ \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_f} \right\} \times \left\{ \frac{\partial z_f}{\partial o_{12}} \right\} \times \left\{ \frac{\partial o_{12}}{\partial z_p} \right\} \times \left\{ \frac{\partial z_p}{\partial w_{12}^1} \right\}$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$-(y-\hat{y}) \quad w_{21}^2 \quad o_{12}(1-o_{12}) \quad x_{i1}$$

$$\frac{\partial L}{\partial w_{12}^1} = -(y-\hat{y}) \cdot w_{21}^2 \cdot o_{12} (1-o_{12}) \cdot x_{i1}$$

$$\frac{\partial L}{\partial w_{22}^1} = -(y-\hat{y}) \cdot w_{21}^2 \cdot o_{12} (1-o_{12}) \cdot x_{i2}$$

$$\frac{\partial L}{\partial b_{12}} = -(y-\hat{y}) \cdot w_{21}^2 \cdot o_{12} (1-o_{12})$$

Backpropagation - why?

for i in range (epochs):

for j in range (X.shape[0]):

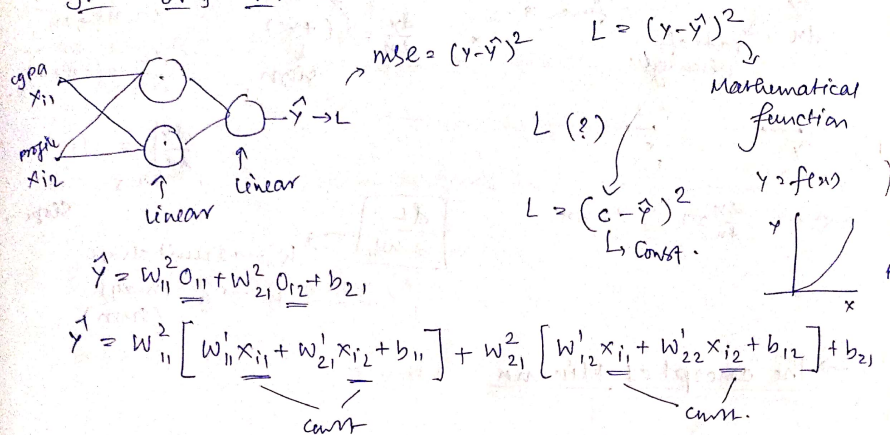
- select 1 row (random)
- predict (using forward propagation) → bce
- calculate loss (using loss-function → mse)
- update weights and biases - GD

$$w_n = w_0 - \eta \cdot \left[\frac{\partial L}{\partial w_0} \right] \rightarrow \text{magic} !!$$

calculate avg loss for the epoch.

→ Loss fn is a function of all trainable parameters

eg: $w_{11}, w_{21}, w_{12}, w_{22}, b_{11}, b_{12}, b_{21}, b_{22}$



$L(\hat{y}) = L(w, b)$

all the weights & biases.

→ Concept of Gradient

Gradient descent.

$y = f(x) = x^2 + x$

$\frac{dy}{dx} = \frac{d}{dx}(f(x)) = \frac{d}{dx}(x^2 + x) = 2x + 1$

derivative

$y \rightarrow x \rightarrow \text{only one parameter} \rightarrow \text{derivative} \cdot \frac{d}{dx}$

dependent

$z = f(x, y) = x^2 + y^2$

$\frac{\partial z}{\partial x} = 2x$

$\frac{\partial z}{\partial y} = 2y$

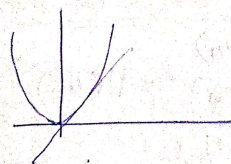
Gradients.

Complex $\rightarrow L(w_{11}^1, w_{12}^1, \dots, w_{21}^2, b_{21}, b_{11}, b_{12})$

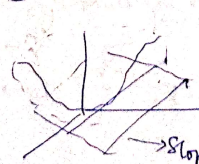
$L(w, b)$

9 parameters.

$y = f(x) = x^2$



$\left[\frac{\partial L}{\partial w}, \frac{\partial L}{\partial b} \right]$



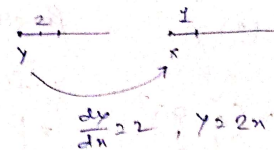
$z = x^2 + y^2$

$z = f(x, y)$

Calculating 9 different slopes w.r.t. each dim.

→ Concept of derivative → derivative at a point.

$\frac{dy}{dx}$ rate of change.



$\frac{dy}{dx} = 2(+ve)$ mag & sign

$y = x^2 + 2x$

$\frac{dy}{dx} = (2x + 1)_{x=2.5}$

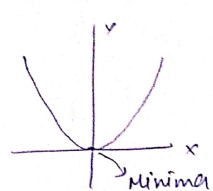
$\frac{dy}{dx} = 11$ Slope.

$\frac{\partial L}{\partial w_{11}}$

$w_{11} \rightarrow \text{unit change}$

$L \rightarrow ?$ (mag) (sign)

→ The concept of Minimum



$\left(\frac{dy}{dx} \right) = 0$

$x = ?$ minima

$y = x^2$

$\frac{dy}{dx} = 2x = 0$

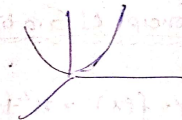
$x \geq 0 \rightarrow \text{Minima}$

$z = x^2 + y^2$

$\frac{\partial z}{\partial x} = 2x = 0 \rightarrow x = 0$

$\frac{\partial z}{\partial y} = 2y = 0 \rightarrow y = 0$

minima $\rightarrow (0, 0)$



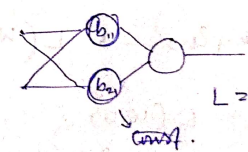
→ Backpropagation intuition

$w_n = w_0 - \eta \frac{\partial L}{\partial w_0}$

$w_n = w_0 - \frac{\partial L}{\partial w_0}$

$\eta = 1$

9 step



$L = L(w, b)$

Const.

$b_{21} = b_{21} - \left[\frac{\partial L}{\partial b_{21}} \right]$

$L = (y - \hat{y})^2$

$\hat{y} \rightarrow (b_{21})$

$L \rightarrow (b_{21}) = L(b_{21})$

$\frac{\partial L}{\partial b_{21}}$

L derivative w.r.t. b_{21}

$\frac{\partial L}{\partial b_{21}} = +ve \rightarrow b_{21} \uparrow L \uparrow$

So we have to reduce b_{21}

So we have to $\uparrow b_{21}$

